

Analyzing Financial Statements

MBA Financial Management & Analysis Topic 5

- 5.1 Definition of Financial Statements Analysis (FS)
- 5.2 The objectives of analyzing FS
- 5.3 The process of Analyzing FS
- 5.4 The source of Information for Analyzing FS
- 5.5 The Analysis



What Is Financial Statement Analysis?

Financial statement analysis is the process of examining and interpreting the information contained in an entity's financial statements to evaluate its financial performance, financial position, cash flows, and overall financial health for informed decision-making.

Financial Performance

How well the organization generates revenue, controls costs, and reports profit over a given period.

Financial Position

The structure of assets, liabilities, and equity at a specific point in time, the balance sheet snapshot.

Cash Flows

The organization's ability to generate and manage cash from operating, investing, and financing activities.

Overall Financial Health

An integrated view combining profitability, liquidity, solvency, and efficiency to support strategic decisions.

Why Analyze Financial Statements?

Financial statement analysis serves multiple stakeholders across a wide range of decisions. The six major reasons are outlined below.

1 Evaluate Organizational Performance

Helps management assess how well the entire organization or individual business units perform in terms of profitability, efficiency, growth, and financial health. Identifies strong and weak areas for corrective action e.g., comparing profits of Nairobi, Mombasa, and Kisumu branches.

2 Credit Analysis for Lending

Banks and lenders analyze financial statements to determine whether a borrower has sufficient income, cash flows, and assets to repay a loan. They assess liquidity, profitability, leverage, and cash flow before approving credit e.g., reviewing current ratio and debt-to-equity ratio before approving a \$2 million loan.

3 Credit Rating for Default Risk

Credit rating agencies evaluate financial statements to estimate the likelihood of default. Strong financial performance generally leads to better credit ratings and lower borrowing costs e.g., a rating upgrade after observing consistent profits and low debt levels.

Why Analyze Financial Statements? (Continued)

1 Investment Decision

Investors analyze financial statements to determine whether an investment is likely to generate satisfactory returns and whether the business has good growth prospects and manageable risks e.g., comparing EPS, ROE, and dividend history of two listed companies before purchasing shares.

2 Valuation in Mergers & Acquisitions (M&A)

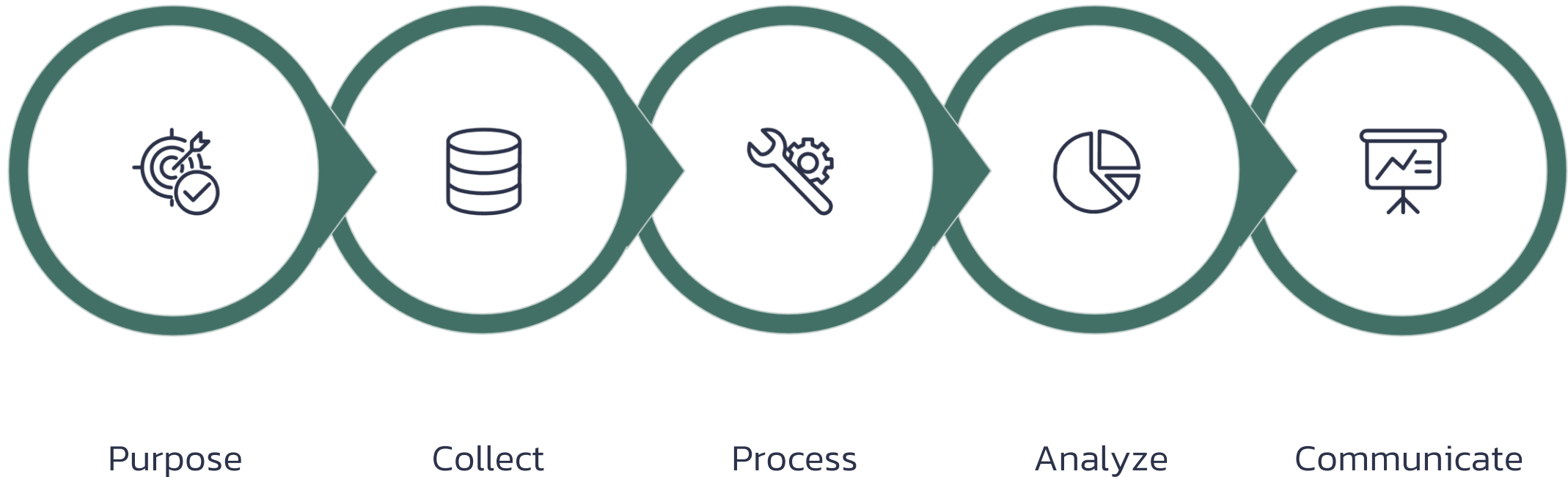
Financial statement analysis helps determine the fair value of a company before it is acquired, merged, or sold. Buyers evaluate assets, liabilities, earnings, cash flows, and future growth potential e.g., Company A analyzes Company B's financials before negotiating an acquisition price of \$150 million.

3 Assess Compliance with Loan Covenants & Regulations

Lenders, regulators, and management analyze financial statements to ensure compliance with loan agreements, accounting standards, tax laws, and industry regulations. Failure to comply may result in penalties, loan default, or regulatory action e.g., a bank verifying a borrower maintains a debt-to-equity ratio below 1.5:1.

The Process of Financial Statement Analysis

Financial statement analysis is a systematic, six-step process that transforms raw accounting data into useful information for decision-making. Each step builds on the previous one.



The process is continuous rather than a one-time exercise analysts must monitor, update, and revise their findings as new information becomes available.

Step 1: Articulate the Purpose and Context

What It Means

Clearly define why the financial statements are being analyzed, who will use the results, and the decisions to be made.

Recall

1. Performance
2. Valuation
3. Trends
4. Forecasting

Key Activities

- Identify the objective of the analysis
- Determine the users investors, lenders, management, regulators
- Understand the industry, economic environment, and reporting period

Practical Example

A bank analyzes a company's financial statements to determine whether to approve a loan, while an investor analyzes them to decide whether to buy shares. The purpose shapes every subsequent step of the analysis.

Steps 2 & 3: Collect and Process the Data

Step 2: Collect Data

Gather all relevant financial and non-financial information needed for the analysis.

- Obtain audited financial statements
- Collect annual reports, notes, budgets, industry reports, and economic data
- Verify that information is complete and reliable

Example: Collect the SFP, SPoL, SCF, SCE, and industry benchmark data for the last five years.

Step 3: Process the Data

Organize and prepare the information so that meaningful comparisons can be made.

- Classify financial data
- Calculate financial ratios
- Prepare common-size statements
- Conduct trend and comparative analyses
- Adjust for unusual or one-off items where necessary

Example: Calculate current ratio, debt-to-equity ratio, gross profit margin, and prepare a five-year sales trend analysis.

Steps 4, 5 & 6: Analyze, Conclude, and Follow Up

1

Analyze and Interpret the Data

Compare current performance with prior years, budgets, competitors, and industry averages. Identify causes of significant changes and assess profitability, liquidity, efficiency, solvency, and cash flows.

Example: Current ratio falls from 2.0 to 1.1, indicating declining liquidity, while profit margins improve due to better cost control.

2

Develop and Communicate Conclusions

Summarize key findings, highlight risks and opportunities, and recommend actions. Present findings through reports, presentations, or dashboards.

Example: Recommend reducing inventory levels, improving debt collection, and postponing major borrowing to strengthen liquidity.

3

Follow Up and Update

Monitor whether recommendations have been implemented and update the analysis as new information becomes available. Review new financial statements, update ratios and forecasts, and revise recommendations where necessary.

Example: Six months later, management reviews updated statements to determine whether liquidity improved after implementing stricter credit control.

Sources of Information

Financial statement analysts use information from both financial and non-financial sources to obtain a complete understanding of a company's financial performance, position, risks, and future prospects. Six major sources are identified.



Source 1: Financial Statements

Primary Reports Under IFRS

Financial statements are the primary reports that provide information about a company's financial position, financial performance, cash flows, and changes in equity.

- Statement of Financial Position (SFP)
- Statement of Profit or Loss (SPoL)
- Statement of Comprehensive Income
- Statement of Cash Flows (SCF)
- Statement of Changes in Equity (SCE)
- Notes to the Financial Statements

Why It Is Useful

Enables analysts to assess profitability, liquidity, solvency, efficiency, and cash flow performance using financial ratios and trend analysis. Financial statements are the foundation of all quantitative analysis and provide the raw data from which ratios, trends, and comparisons are derived.

Sources 2 & 3: MD&A and Audit Report

Management Discussion & Analysis (MD&A)

A narrative report prepared by management explaining the company's financial results, business performance, major risks, opportunities, and future outlook. It helps users understand the reasons behind reported financial results.

Information obtained: Explanation of revenue growth, cost increases, business strategy, market conditions, risks, future plans, and expected performance.

Why useful: Helps analysts understand management's perspective, business strategy, and the factors influencing financial performance and future prospects.

Audit Report

An independent auditor's report expressing an opinion on whether the financial statements present a true and fair view in accordance with the applicable financial reporting framework.

Information obtained: Unmodified (clean) opinion, qualified opinion, adverse opinion, disclaimer of opinion, key audit matters, and emphasis of matter paragraphs.

Why useful: Increases confidence in the reliability of the financial statements and highlights significant audit issues or uncertainties that may affect decision-making.

Sources 4, 5 & 6: Regulatory, Press & Website



Returns/Reports to Regulators

Reports submitted to government agencies to comply with legal and industry-specific requirements. Includes tax returns, prudential returns to central banks, stock exchange filings, insurance solvency reports, capital adequacy reports, and statutory returns. Enables assessment of regulatory compliance, financial stability, and industry-specific risks.



Corporate Reports to the Press

Public announcements issued through newspapers, press releases, media interviews, or investor briefings. Includes quarterly earnings announcements, M&A news, new products, leadership changes, major contracts, dividends, and restructuring plans. Provides timely information about significant events influencing performance, market value, or reputation.



Company Website

Information published on the company's official website complementing the financial statements. Includes annual reports, investor presentations, sustainability reports, corporate governance reports, press releases, financial calendars, and strategic plans. Provides current and comprehensive information about operations, governance, and investor relations.

Sources of Information: Summary



Financial Statements

SFP, SPoL, SCF, SCE, and Notes the primary source for all quantitative ratio analysis.



MD&A

Management's narrative explaining results, strategy, risks, and future outlook.



Audit Report

Independent opinion on reliability of financial statements; highlights key audit matters.



Regulatory Returns

Tax returns, prudential returns, stock exchange filings, and statutory reports.



Press Releases

Quarterly earnings, M&A announcements, dividends, leadership changes, and restructuring news.



Company Website

Annual reports, investor presentations, sustainability reports, governance disclosures, and strategic plans.

Approaches to Analyzing Financial Statements

Two broad approaches are used in analyzing financial statements: **Absolute Analysis** and **Relative Analysis**. Each serves a different purpose and has distinct strengths and limitations.

Absolute Analysis

Uses actual financial statement figures. Examines whether a profit was made, how figures compare with prior years (trend), or with other organizations in the same industry (cross-section). Also involves comparing changes in values of assets, liabilities, and capital from one period to the next.

Limitation: Good as a starting point but interpretation can be misleading because it ignores the size of the organization.

Relative Analysis

Financial statements are standardized using ratios. Removes the size distortion inherent in absolute analysis. Two main sub-approaches: **Common Size Analysis** (horizontal and vertical) and **Categorical Analysis** (computing financial ratios grouped into categories).

Advantage: Enables meaningful comparisons across organizations of different sizes and across different time periods.

Common Size Analysis: Horizontal vs. Vertical

Horizontal Analysis

Takes the previous period's financial statements as the base. Subsequent period items are expressed as a percentage of the previous period's figures. This reveals the rate of increase or decrease over time.

Horizontal analysis is very popular and the most commonly used for presenting trend analysis — comparing the same company over time.

Vertical Analysis

A key item is identified and all other items are expressed as a percentage of that item:

- **SPoL:** Revenue is the key item, all other incomes and expenses expressed as a % of revenue
- **SFP:** Total assets is the key item, all other items expressed as a % of total assets
- **SCF:** Total receipts and total payments determined, items expressed as a % of each total

Categorical Analysis: The Six Categories

The more common method of analyzing financial statements is computing financial ratios across the various financial statements and grouping them into six categories also referred to as the **financial health check** for an organization.



1. Efficiency

Ability to generate revenue from available resources



2. Profitability

Ability to manage costs and report a profit



3. Cash Flow

Ability to generate cash from operations



4. Liquidity

Ability to pay short-term obligations



5. Solvency/Gearing

Ability to settle total liabilities from total assets



6. Investor Value

Ability to generate value for owners

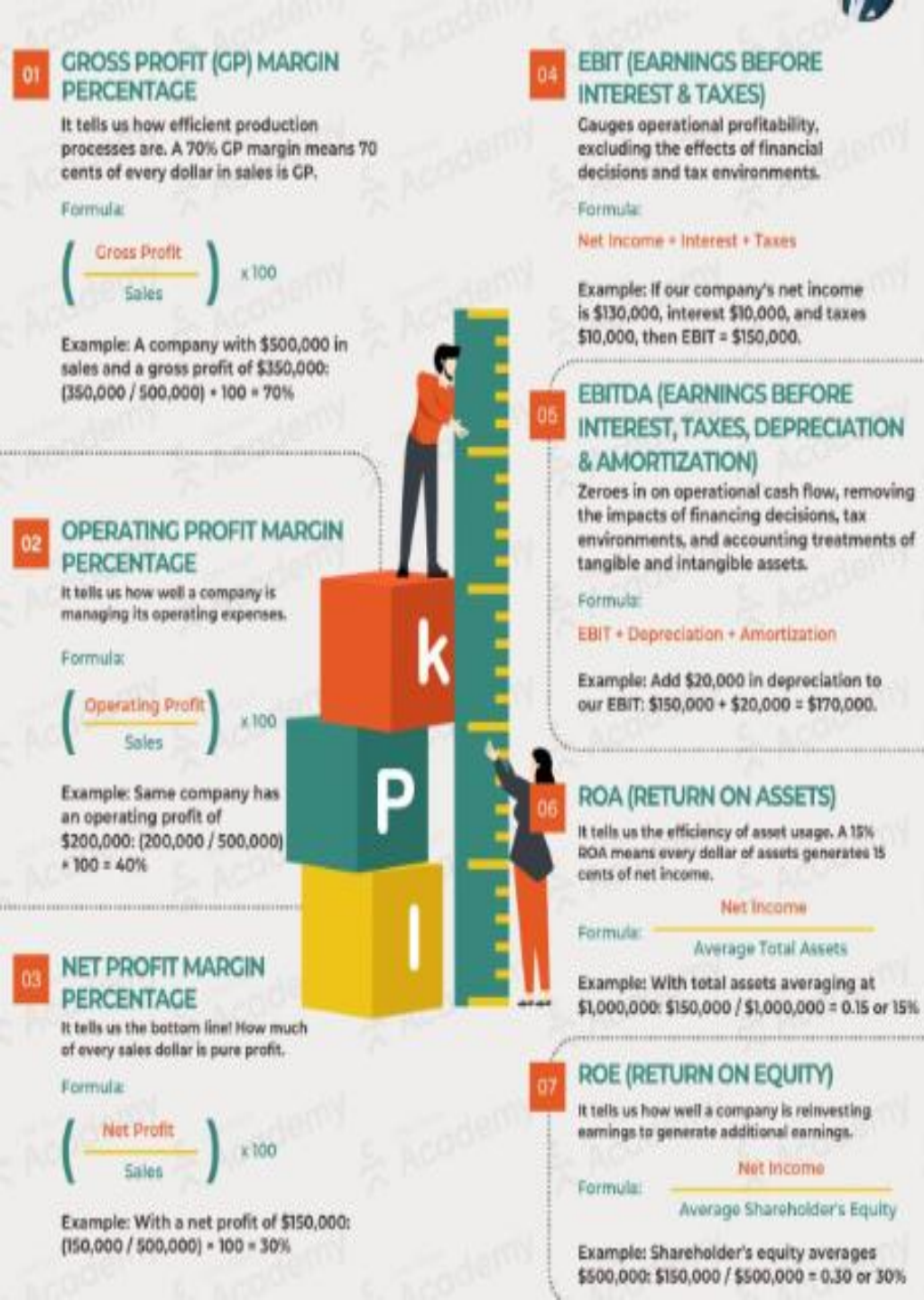


Category 1: Efficiency Ratios

Efficiency (also called Activity) measures how efficiently the organization generates revenue from its resources. Key questions: Is the organization selling goods quickly? Collecting cash from credit customers promptly? Paying credit suppliers efficiently? Generating revenue from long-term assets?

Efficiency Ratios: Formulas & Interpretation

Ratio	Formula	Explanation	Remark
Payables Turnover (Days of Purchases — DOP)	$\text{Credit Purchases} \div \text{Avg/Closing Payables}; \text{Days} = 365 \div \text{Turnover}$	Number of times the firm makes credit purchases in a year; how long it takes to pay credit suppliers	Higher turnover = more efficient; fewer days = better
Inventory Turnover (Days of Inventory — DOI)	$\text{Cost of Sales} \div \text{Avg/Closing Inventory}; \text{Days} = 365 \div \text{Turnover}$	Number of times inventory is purchased and sold in a year; how long it takes to sell inventory	Higher turnover = more efficient; fewer days = better
Receivables Turnover (Days of Sales — DOS)	$\text{Credit Sales} \div \text{Avg/Closing Receivables}; \text{Days} = 365 \div \text{Turnover}$	Number of times the firm generates credit sales; how long it takes to collect cash from credit customers	Higher turnover = more efficient; fewer days = better
Noncurrent Assets Turnover	$\text{Total Revenue} \div \text{Noncurrent Assets}$	Ability to generate revenue from noncurrent assets	Higher = better (rarely used)
Total Assets Turnover	$\text{Total Revenue} \div \text{Total Assets}$	Ability to generate revenue from all resources	Higher = better; signifies efficient use of resources



Category 2: Profitability Ratios

Profitability ratios measure how profitable the organization is. Profitability is assessed at three levels: in relation to **revenue**, in relation to **resources (total assets)**, and in relation to **capital (equity)**. Most ratios in this category are expressed as percentages the higher the ratio, the better.

Profitability Ratios: Formulas & Interpretation

Ratio	Formula	Explanation	Remark
Gross Profit Margin %	$\text{Gross Profit} \div \text{Revenue}$	How much gross profit is made for every unit of sale	Higher = better
Operating Profit Margin %	$\text{PBIT} \div \text{Revenue}$	How profitable the organization is in controlling operating expenses	Higher = better
Net Profit Margin % (Return on Sales)	$\text{PAT} \div \text{Revenue}$	Net profit for every unit of sale	Higher = better
Return on Assets/Investment (ROI) %	$\text{PAT} \div \text{Avg/Total Assets}$	Return (net profit) per unit of asset — how profitable the assets are	Higher = better
Return on Equity (ROE) %	$\text{PAT} \div \text{Avg/Total Equity}$	Return per unit of capital contributed by owners; profitability from owners' perspective	Higher = better
Return on Capital Employed (ROCE) %	$\text{PBIT} \div (\text{Equity} + \text{Noncurrent Liabilities})$	Return per unit of contribution from long-term lenders	Higher = better; not commonly used currently

Category 3: Cash Flow Ratios

Cash flow ratios measure the ability of the organization to generate cash and evaluate its efficiency in using that cash. This is a relatively recent category and may not be found in many older accounting texts. The main unit of analysis is **Cash Flows from Operating Activities (CFO)**, assessed in relation to revenue, total assets, equity, and liabilities.

Ratio	Formula	Explanation	Remark
Cash Flow-to-Revenue	$\text{CFO} \div \text{Net Revenue}$	Operating cash flow generated for each dollar of revenue	Higher = better
Cash Return-on-Assets	$\text{CFO} \div \text{Avg Total Assets}$	Cash generated from operations for every unit of resource	Higher = better
Cash Return-on-Equity	$\text{CFO} \div \text{Avg Total Equity}$	Cash generated from operations for every unit of owners' capital	Higher = better
Cash Flow Per Share	$(\text{CFO} - \text{Preference Dividends}) \div \text{Weighted Avg Ordinary Shares}$	Variation of basic EPS using CFO instead of net income	Higher = better
Debt Coverage Ratio	$\text{CFO} \div \text{Total Liabilities}$	Extent to which CFO can pay all liabilities	Higher = better
Debt Payment Ratio	$\text{CFO} \div \text{Long-term Debt (Noncurrent Liabilities)}$	Firm's ability to satisfy long-term debt with operating cash flow	Higher = better

Category 4: Liquidity Ratios

Liquidity measures the ability of the organization to meet its short-term maturing obligations (current liabilities). It is expected that current assets will be used to pay current liabilities. The ability to pay current liabilities is measured at three levels:



Level 1 — Current Assets vs. Current Liabilities

Can the organization pay current liabilities from all current assets? If current assets exceed current liabilities, the firm is considered liquid.

Current Ratio = Current Assets ÷ Current Liabilities. Benchmark: approximately 2:1. Higher = better.



Level 2 — Excluding Inventory

Can the organization pay current liabilities without having to sell inventory (which may not be sold quickly)?

Acid Test Ratio = (Current Assets – Inventory) ÷ Current Liabilities. Benchmark: approximately 1:1. Higher = better.



Level 3 — Cash Only

Can the organization pay current liabilities from cash in hand, cash at bank, and short-term financial assets?

Cash Ratio = (Cash + Short-term Financial Assets) ÷ Current Liabilities. Benchmark: approximately 0.5. Higher = better.

Category 5: Gearing / Solvency / Leverage Ratios

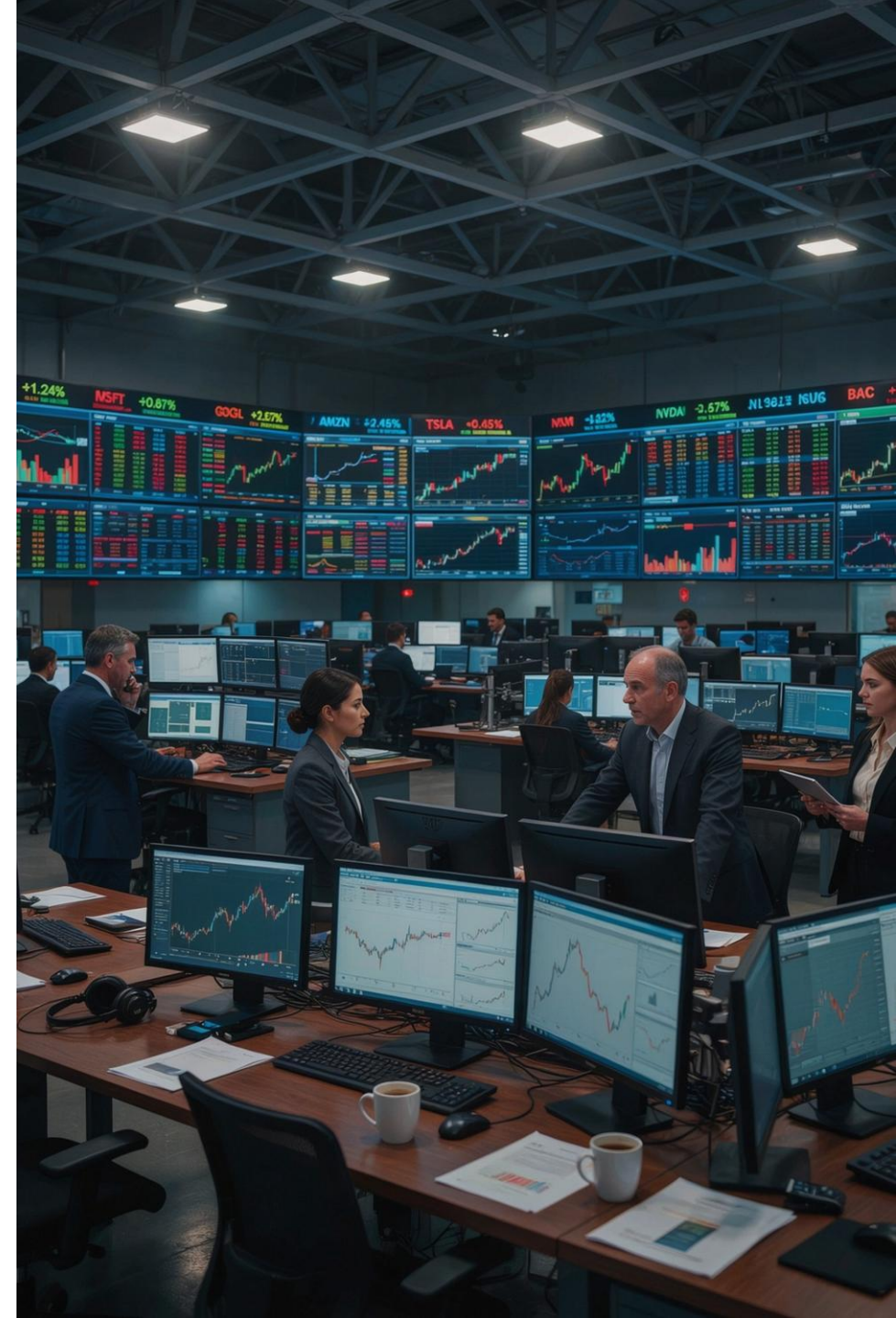
These ratios measure the ability of the organization to meet its short-term and long-term maturing obligations i.e., total liabilities. As a rule, total assets should exceed total liabilities, indicating the organization can settle liabilities from its assets (solvency). If total liabilities exceed total assets, the organization is **insolvent**. Apart from the interest coverage ratio, the lower the ratios the better.

Ratio	Formula	Explanation	Remark
Debt Ratio %	$\text{Total Liabilities} \div \text{Total Assets}$	Proportion (%) of assets funded by liabilities	Lower = better
Debt-to-Equity Ratio	$\text{Total Liabilities} \div \text{Total Equity}$	Ratio of contribution by lenders to contribution by owners	Lower = better
Interest Coverage	$\text{PBIT} \div \text{Interest Expense}$	How many times interest expense can be paid from profit	Higher = better

📌 Note: Some textbooks use only long-term loans in the numerator for Debt Ratio and Debt-to-Equity. In this course, we use total liabilities.

Category 6: Investor / Owner Value Ratios

These ratios measure the value of the organization according to investors or owners. Key concerns for owners include: Is the organization generating sufficient profits? Is it paying a reasonable cash return (dividends)? Is it increasing owners' capital through retained profits or rising market price?



Investor Ratios: Formulas & Interpretation

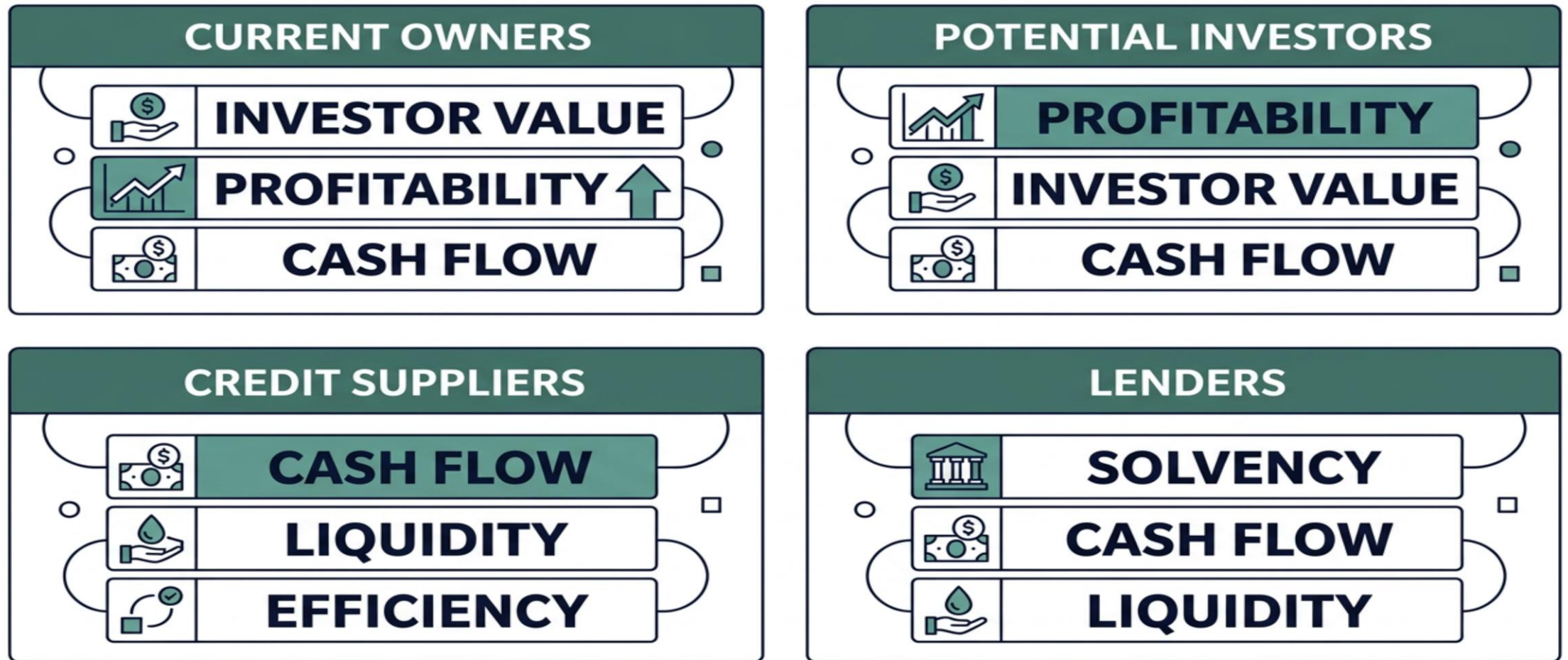
Ratio	Formula	Explanation	Remark
Earnings Per Share (EPS)	$(\text{PAT} - \text{Preference Dividends}) \div \text{No. of Ordinary Shares}$	Profit on a per-share basis — how much profit is made per share	Higher = better
Dividend Per Share (DPS)	$\text{Total Ordinary Dividends} \div \text{No. of Ordinary Shares}$	Actual cash an investor expects from each share	Higher = better
Book Value Per Share (BPS)	$\text{Total Shareholders' Funds (excl. pref. shares)} \div \text{No. of Ordinary Shares}$	Accounting value of a share given current valuation of assets and liabilities	Higher = better
Earnings Yield %	$\text{EPS} \div \text{Market Price}$	Total return (profit) an investor gets for every unit of investment at current market price	Higher = better
Dividend Yield %	$\text{DPS} \div \text{Market Price}$	Cash return only — same as earnings yield but limited to dividends paid	Higher = better
Price-Earnings (P/E) Ratio	$\text{Price Per Share} \div \text{EPS}$	Amount an investor is willing to pay for every unit of profit	High P/E = high growth potential ("growth companies")

Priority Areas of Analysis for Different Users

Different users of financial statements emphasize different aspects of analysis depending on their objectives. The table below shows the priority order from highest to lowest for each major user group.

User Group	Priority Order (Highest → Lowest)	Reason
Current Owners / Investors	Investor Value → Profitability → Cash Flow → Solvency → Liquidity → Efficiency	Focus on maximizing shareholder wealth while ensuring sustainable financial performance
Potential Investors	Profitability → Investor Value → Cash Flow → Solvency → Liquidity → Efficiency	Interested in future returns, growth prospects, and financial strength before investing
Credit Suppliers (Trade Creditors)	Cash Flow → Liquidity → Efficiency → Solvency → Profitability → Investor Value	Primarily concerned with the firm's ability to pay short-term trade obligations promptly
Lenders / Financiers	Solvency → Cash Flow → Liquidity → Profitability → Efficiency → Investor Value	Focus on the borrower's long-term financial stability and capacity to repay principal and interest

User Priorities at a Glance



Understanding user priorities helps analysts tailor their reports and focus on the metrics most relevant to the decision at hand. A lender and an investor looking at the same company will draw very different conclusions from the same financial statements.

Application Areas for Financial Ratios

Financial ratios are not only used to evaluate past performance. They extend to comparing businesses, estimating missing financial information, and preparing future financial projections. Four major application areas are identified.



Application 1: Trend Analysis (Time-Series)

What It Is

Evaluating a company's financial performance over several consecutive accounting periods by calculating ratios for each year and analyzing whether performance is improving, stable, or declining.

Practical Example

A company's Gross Profit Margin increased from 28% (2023) to 31% (2024) and 35% (2025), indicating improved cost control and pricing strategies over three years.

Benefits to Users

- Identifies long-term trends in financial performance
- Measures progress toward strategic goals
- Detects early warning signs of deterioration
- Supports strategic planning and forecasting
- Provides context for current-year results

How Ratios Are Applied

Financial ratios are calculated for consecutive years and compared to identify patterns. Analysts look for consistent improvement, sudden deterioration, or cyclical fluctuations that require explanation.

Application 2: Cross-Sectional Analysis (Benchmarking)

What It Is

Comparing one company's financial performance with competitors or industry averages during the same period. Ratios are compared against competitors, industry averages, or best-performing firms to evaluate relative performance.

Practical Example

Company A has a Current Ratio of 2.0, while the industry average is 1.5. This suggests Company A has stronger liquidity than most competitors in its sector.

Benefits to Users

- Helps identify relative strengths and weaknesses
- Supports investment decisions by comparing alternatives
- Encourages adoption of best practices from industry leaders
- Provides context that time-series analysis alone cannot offer
- Highlights competitive positioning

Application 3: Preparing Statements with Incomplete Information

Using known ratios and partial financial information to estimate missing figures in financial statements. Rearranging ratio formulas allows accountants and analysts to compute unknown values such as sales, inventory, current liabilities, or total assets.

Example 1 — Finding Current Liabilities

If Current Ratio = 2:1 and Current Assets = Sh 40 million, then:

$$\text{Current Liabilities} = \text{Current Assets} \div \text{Current Ratio} \\ = 40 \div 2 = \text{Sh 20 million}$$

Example 2 — Finding Sales

If Gross Profit Margin = 25% and Gross Profit = Sh 15 million, then:

$$\text{Sales} = \text{Gross Profit} \div \text{GP Margin} = 15 \div 0.25 = \text{Sh 60 million}$$

This application is particularly useful in accounting examinations, auditing, business valuation, forensic accounting, and financial analysis where complete records are unavailable.

Application 4: Preparing Forecast (Pro Forma) Financial Statements

What It Is

Using historical ratios and expected future assumptions to project future financial performance and position. Historical financial ratios are combined with projected sales, growth rates, and financing assumptions to estimate future income statements and balance sheets.

Practical Example

A company expects sales to increase by 12% next year. If the Net Profit Margin has historically averaged 10%, projected net profit is estimated at 10% of forecast sales. Similarly, historical Asset Turnover and Debt Ratios help forecast future assets and financing needs.

Benefits

- Supports budgeting and business planning
- Assists with investment appraisal and loan applications
- Facilitates capital raising and strategic decision-making

- ❏ Overall, financial ratios are powerful analytical tools that extend beyond evaluating historical performance enabling stakeholders to monitor trends, benchmark against peers, reconstruct incomplete statements, and prepare realistic financial forecasts.

Limitations of Financial Ratio Analysis

While financial ratios are powerful tools, they have significant limitations that analysts must be aware of. Fifteen key limitations are identified below.



Limitations 1–5: Data, Policies & Manipulation

1. Based on Historical Data

Ratios are derived from past financial statements and may not reflect current or future business conditions. A company's past performance is not necessarily a reliable predictor of future results.

2. Different Accounting Policies

Variations in accounting methods such as depreciation methods, inventory valuation (FIFO vs. weighted average), and revenue recognition reduce comparability between companies.

3. Inflation Effects

Financial statements prepared using historical costs may not reflect the current value of assets and liabilities, making ratios misleading during periods of significant inflation.

4. Window Dressing

Management may temporarily alter financial figures e.g., delaying purchases or accelerating collections at year-end to improve reported ratios and present a more favorable picture.

5. Industry Differences

Ratios vary significantly across industries, making comparisons between businesses in different sectors inappropriate. A high debt ratio may be normal in banking but alarming in manufacturing.

Limitations 6–10: Seasonal, Qualitative & Economic Factors

6. Seasonal Business Fluctuations

Ratios calculated at year-end may not represent the company's average financial position throughout the year, particularly for businesses with strong seasonal patterns.

7. Ignores Qualitative Factors

Ratios do not capture important non-financial factors such as management quality, employee skills, customer satisfaction, innovation, or brand reputation all of which significantly influence performance.

8. No Universal Benchmarks

There are no ideal ratios applicable to all businesses. Acceptable levels differ by industry, size, and business model, making it difficult to determine what constitutes a "good" ratio.

9. Influenced by Economic Conditions

Inflation, recession, interest rates, exchange rates, and government policies can distort ratio interpretation, making it difficult to isolate management performance from external factors.

10. Ratios Alone Do Not Explain Causes

Ratios identify symptoms of performance problems but do not explain why changes occurred. Additional investigation is always required to understand the underlying causes.

Limitations 11–15: Accuracy, Size & Strategy

11. Dependence on Accurate Financial Statements

Incorrect, incomplete, or fraudulent financial statements produce unreliable ratios. The quality of ratio analysis is only as good as the quality of the underlying financial data.

12. Different Company Sizes

Comparing large and small firms using ratios alone may overlook differences in economies of scale, market power, and available resources that ratios cannot fully capture.

13. One Ratio Gives an Incomplete Picture

A single ratio cannot adequately assess overall financial performance. Several ratios across multiple categories should always be analyzed together for a balanced view.

14. Changes in Business Strategy

Expansion, restructuring, mergers, or new business models can make historical ratios less meaningful, as the business being analyzed today may be fundamentally different from the business of prior years.

15. Off-Balance Sheet Items

Some obligations, such as operating lease commitments, guarantees, and contingent liabilities may not be fully reflected in the financial statements, leading to misleading solvency and leverage ratios.

Useful Tips When Using Ratios

1 Use Ratios Together with Full Financial Statements

Ratios alone provide limited information. Always use them alongside the complete financial statements ; SFP, SPoL, SCF, and SCE, for a clear and comprehensive picture of the organization's financial health.

3 Consider Qualitative Aspects

Always consider the nature of the organization, the industry it operates in, and the general economic environment. Ratios provide important but limited information. Advanced techniques such as sensitivity and scenario analysis may also be employed.

2 Use Several Categories of Ratios

Analyze ratios across multiple categories efficiency, profitability, cash flow, liquidity, solvency, and investor value and be cautious when drawing conclusions from any single ratio or category.

4 Be Aware of Sector-Specific Ratios

Certain sectors such as banking and insurance have their own specialized ratios (e.g., capital adequacy ratio, combined ratio). Students are encouraged to explore sector-specific ratios beyond the core scope covered here.

Efficiency Ratios: Summary Recap

A quick reference summary of the efficiency (activity) ratios covered in this topic, with their key interpretation guidance.

Payables Turnover / DOP

Formula: $\text{Credit Purchases} \div \text{Avg Payables}$

Days: $365 \div \text{Turnover}$

Measures how quickly the firm pays its credit suppliers.
Fewer days = more efficient payment management.

Inventory Turnover / DOI

Formula: $\text{Cost of Sales} \div \text{Avg Inventory}$

Days: $365 \div \text{Turnover}$

Measures how quickly inventory is sold. Fewer days =
faster inventory movement = more efficient operations.

Receivables Turnover / DOS

Formula: $\text{Credit Sales} \div \text{Avg Receivables}$

Days: $365 \div \text{Turnover}$

Measures how quickly cash is collected from credit customers. Fewer days = faster collections = better liquidity.

Total Assets Turnover

Formula: $\text{Total Revenue} \div \text{Total Assets}$

Measures how efficiently all resources generate revenue.
Higher = better use of assets.

Profitability Ratios: Summary Recap

Gross Profit Margin

$\text{Gross Profit} \div \text{Revenue}$

Measures profitability after direct costs. Reflects pricing power and cost of goods sold management.

Operating Profit Margin

$\text{PBIT} \div \text{Revenue}$

Measures profitability after operating expenses. Reflects operational efficiency and overhead control.

Net Profit Margin

$\text{PAT} \div \text{Revenue}$

Bottom-line profitability after all costs including interest and tax.

ROI (Return on Assets)

$\text{PAT} \div \text{Total Assets}$

How profitably the firm uses its total asset base to generate net profit.

ROE (Return on Equity)

$\text{PAT} \div \text{Total Equity}$

Profitability from the owners' perspective — return per unit of equity invested.

ROCE

$\text{PBIT} \div (\text{Equity} + \text{NCL})$

Return per unit of long-term capital employed. Less commonly used in current practice.

Liquidity & Solvency: Summary Recap

Liquidity Ratios

Current Ratio

Current Assets ÷
Current Liabilities.
Benchmark ~2:1.
Higher = better.

Acid Test Ratio

(Current Assets –
Inventory) ÷ Current
Liabilities. Benchmark
~1:1. Higher = better.

Cash Ratio

(Cash + Short-term Financial Assets) ÷ Current
Liabilities. Benchmark ~0.5. Higher = better.

Solvency / Gearing Ratios

Debt Ratio

Total Liabilities ÷ Total
Assets. Measures % of
assets funded by
liabilities. Lower =
better.

Debt-to-Equity

Total Liabilities ÷ Total
Equity. Ratio of lender
to owner contribution.
Lower = better.

Interest Coverage

PBIT ÷ Interest Expense. Times interest is covered
by profit. Higher = better.

Investor Ratios: Summary Recap

EPS

$(\text{PAT} - \text{Pref. Dividends}) \div \text{Ordinary Shares}$

Profit per share. Higher = better.

DPS

$\text{Total Ordinary Dividends} \div \text{Ordinary Shares}$

Cash return per share. Higher = better.

BPS

$\text{Shareholders' Funds} \div \text{Ordinary Shares}$

Accounting value per share. Higher = better.

Earnings Yield

$\text{EPS} \div \text{Market Price}$

Total return per unit of investment. Higher = better.

Dividend Yield

$\text{DPS} \div \text{Market Price}$

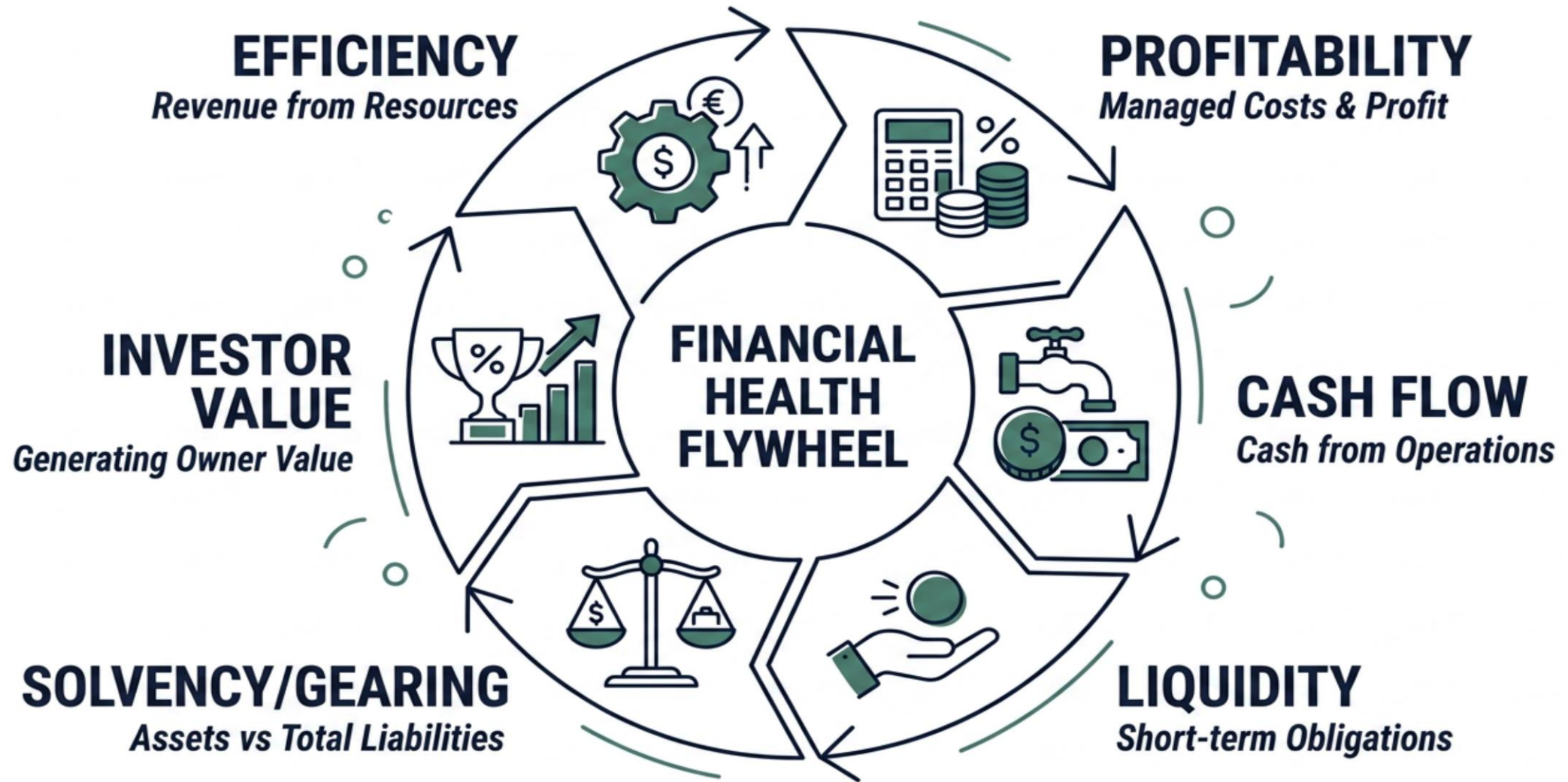
Cash return per unit of investment. Higher = better.

P/E Ratio

$\text{Price Per Share} \div \text{EPS}$

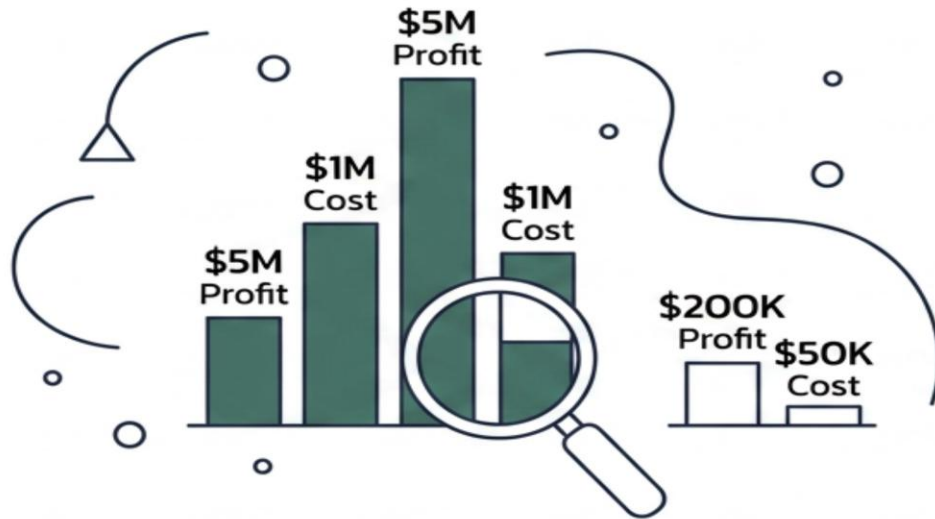
Amount paid per unit of profit. High P/E = growth company.

The Six Financial Health Categories: Full Overview



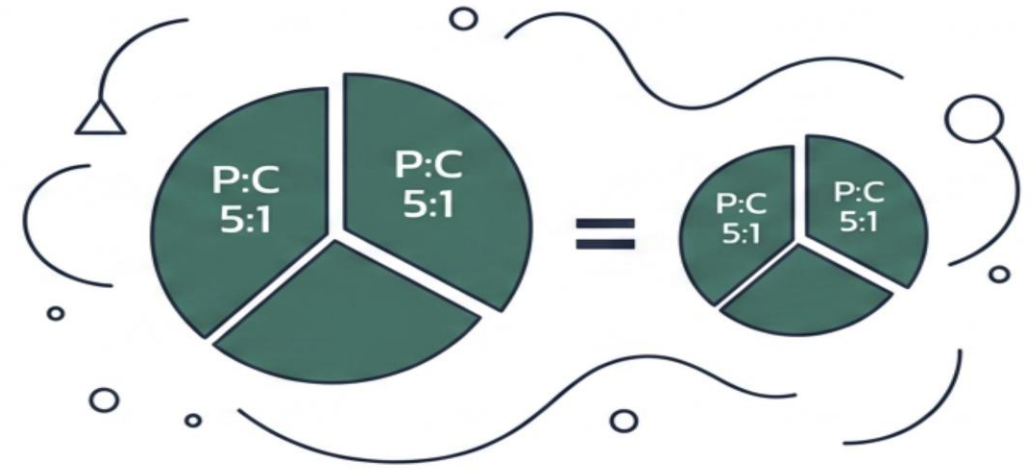
Absolute vs. Relative Analysis: Comparison

ABSOLUTE ANALYSIS



- ✓ Uses actual figures
- ✓ Compares profit made
- ✓ Good starting point
- ✗ Misleading due to size differences

RELATIVE ANALYSIS



- ✓ Standardizes using ratios
- ✓ Ratios remove size distortion
- ✓ Cross-section with industry
- ✓ Horizontal & Vertical Analysis

Both approaches are complementary. Absolute analysis provides the raw picture; relative analysis standardizes it for meaningful comparison across time and across organizations of different sizes.

Applications of Financial Ratios: Summary

Trend Analysis

Calculate ratios over consecutive years to identify improving, stable, or declining performance. Detects early warning signs and supports strategic planning.

Cross-Sectional Analysis

Compare ratios against competitors and industry averages to evaluate relative performance and identify best practices.

Incomplete Information

Rearrange ratio formulas to estimate missing financial figures — useful in auditing, forensic accounting, and examinations.

Pro Forma Forecasting

Combine historical ratios with projected sales and growth assumptions to estimate future financial statements for planning and capital raising.

Limitations of Ratio Analysis: Summary

1. DATA QUALITY



Historical Data



Dependence on Accurate Statements



Off-Balance Sheet Items



Window Dressing

2. COMPARABILITY ISSUES



Different Accounting Policies



Industry Differences



Different Company Sizes



No Universal Benchmarks

3. CONTEXTUAL FACTORS



Inflation Effects



Seasonal Fluctuations



Economic Conditions



Changes in Business Strategy

4. INTERPRETIVE GAPS



Ignores Qualitative Factors



Ratios Do Not Explain Causes



One Ratio Gives Incomplete Picture

Awareness of these limitations is essential for any analyst. Ratios must always be interpreted with caution, in context, and alongside other qualitative and quantitative information.

Topic 5 Summary: Key Takeaways

1 Definition & Purpose

Financial statement analysis is the systematic process of examining financial statements to evaluate performance, position, cash flows, and financial health for informed decision-making across six major use cases.

2 Six-Step Process

Articulate purpose → Collect data → Process data → Analyze & interpret → Communicate conclusions → Follow up. Analysis is continuous, not a one-time exercise.

3 Two Broad Approaches

Absolute analysis uses actual figures (good starting point but size-sensitive); relative analysis standardizes using ratios through common-size analysis (horizontal/vertical) and categorical ratio analysis.

4 Six Health Categories

Efficiency, Profitability, Cash Flow, Liquidity, Solvency/Gearing, and Investor Value together forming the complete financial health check. Different users prioritize these categories differently.

5 Applications & Limitations

Ratios support trend analysis, benchmarking, reconstruction of incomplete statements, and pro forma forecasting but must be used with awareness of 15 key limitations including historical bias, accounting policy differences, and qualitative blind spots.